

# Assembly of the two-loop MHV integrand from maximally positive mutual numerators at five and six points

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## Abstract

For the two-loop four-point MHV amplituhedron, the canonical form on each positive chart is known to be a pure dlog form times the ratio  $D^+/D$ , where  $D = \langle ABCD \rangle$  is the mutual positivity condition and  $D^+$  is the sum of its positive monomials [1]. We test, by exact computation, whether this “maximally positive numerator” structure *assembles* at higher multiplicity: whether the full two-loop MHV integrand at  $n = 5, 6$  equals the sum over all composite cells (ordered pairs of one-loop sign-flip cells) of  $\text{Kermit} \times \text{Kermit} \times D^+/D$ , with no corrections. It does. At  $n = 5$  (9 composite cells) and  $n = 6$  (36 composite cells) the assembly reproduces the local double-pentagon representation of the integrand [2] with ratio exactly 1 in rational arithmetic at independent generic kinematic points; interior seams cancel pairwise; and all well-defined transverse octa-cut residues at  $n = 5$  are quantized as required. Two structural observations explain why the preconditions cannot fail at any  $n$ : in the sign-flip charts,  $\langle ABCD \rangle$  is automatically multilinear in the eight chart variables, and every monomial coefficient is a single Plücker bracket, so the sign split defining  $D^+$  is independent of the choice of positive kinematics. In particular the quasi-linear complication anticipated in [1] does not arise for two-loop MHV in these raw charts at  $n \leq 6$ , suggesting the naive assembly may hold for all  $n$ . All computations are exact and reproducible from the public repository.

## 1 Setup

We use momentum twistors  $Z_1, \dots, Z_n$  on a moment curve (all ordered Plücker brackets positive) and the one-loop MHV amplituhedron in its sign-flip description [1, 3, 4]: a loop line  $AB$  with  $\langle AB i i+1 \rangle > 0$  (twisted cyclic) and exactly two sign flips in the sequence  $\langle AB 1 i \rangle$ ,  $i = 2, \dots, n$ . The flip positions  $(i, j)$ ,  $2 \leq i < j \leq n-1$ , label  $(n-2)(n-3)/2$  cells, each with the positive chart

$$A = Z_1 + x Z_i + w Z_{i+1}, \quad B = -Z_1 + y Z_j + z Z_{j+1}, \quad x, w, y, z > 0. \quad (1)$$

On each cell the canonical form is the corresponding “Kermit” term [2],

$$K_{ij}(A, B) = \frac{\langle AB (1 i i+1) \cap (1 j j+1) \rangle^2}{\langle AB 1 i \rangle \langle AB i i+1 \rangle \langle AB 1 i+1 \rangle \langle AB 1 j \rangle \langle AB j j+1 \rangle \langle AB 1 j+1 \rangle}, \quad (2)$$

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which we verify to be an exact dlog form on its chart ( $K_{ij} \langle AB \ i \ i+1 \rangle \langle AB \ j \ j+1 \rangle x y z = 1$  identically) for all cells at  $n = 5, 6$ .

At two loops, the geometry is the set of ordered pairs of lines  $(AB, CD)$ , each in the one-loop region, with the single mutual condition  $\langle ABCD \rangle > 0$  [1, 4, 5]. Composite cells are ordered pairs of one-loop cells; in each composite chart write  $D = \langle ABCD \rangle$  as a polynomial in the eight chart variables and let  $D^+$  denote the sum of its positive-coefficient monomials.

**Two structural facts.** (i)  $D$  is automatically multilinear: each chart variable enters exactly one column of the  $4 \times 4$  determinant. (ii) Each monomial coefficient is a single Plücker bracket  $\pm \langle Z_a Z_b Z_c Z_d \rangle$  (one basis vector chosen per column), so its sign is fixed by the ordering of the moment curve alone: the split  $D = D^+ - D^-$  is independent of the choice of positive  $Z$ 's. We verified both facts computationally for all 9 composite cells at  $n = 5$  and all 36 at  $n = 6$  (monomial counts ranging from 8 to 26), in addition to their one-line proofs. Consequently the “maximally positive numerator” is well defined at every  $n$ , and the only question is whether the assembly reproduces the integrand.

## 2 The assembly and its tests

**Postulate.** With all quantities evaluated at common representatives of the loop lines,

$$M_2^{(n)} = \sum_{\text{composite cells}} K_{c_1}(A, B) K_{c_2}(C, D) \frac{D_{c_1 c_2}^+}{D_{c_1 c_2}}, \quad (3)$$

where  $D_{c_1 c_2}$ ,  $D_{c_1 c_2}^+$  are evaluated by canonically extracting each pair's chart coordinates from the lines, and the sum is over *ordered* pairs (so  $M_2 = 2 \mathcal{A}^{2\text{-loop}}$  in the convention where the physical amplitude carries 1/2 for indistinguishable loops). At  $n = 4$  this is the known result [1], which our implementation reproduces exactly.

**Gold standard.** We compare (3) against the local double-pentagon representation [2],  $\mathcal{A}_{\text{MHV}}^{2\text{-loop}} = \frac{1}{2} \sum_{i < j < k < l < i} I_{ijkl}$ , with the chiral numerators  $\langle AB \ (i-1 \ i \ i+1) \cap (j-1 \ j \ j+1) \rangle \langle ijkl \rangle \langle CD \ (k-1 \ k \ k+1) \cap (l-1 \ l \ l+1) \rangle$ , implemented verbatim from the reference. In exact rational arithmetic at independent generic kinematic points:

| $n$ | composite cells | double pentagons | ratio $M_2 / \sum I$ |
|-----|-----------------|------------------|----------------------|
| 5   | 9               | 20               | 1 (three points)     |
| 6   | 36              | 60               | 1 (two points)       |

The ratios are exact rational numbers, not floating-point agreements.

**Seams.** Individual composite-cell terms have poles on the spurious hyperplanes  $\langle AB1i \rangle$  ( $i$  non-adjacent). Along exact one-parameter rational paths onto each seam at  $n = 5$ , individual terms diverge as  $1/\varepsilon$  while the assembled sum is constant to six digits across four orders of magnitude in  $\varepsilon$ , and the sum diverges as required on the physical mutual pole  $\langle ABCD \rangle \rightarrow 0$ .

**Leading singularities ( $n = 5$ ).** On the two standard octa-cut families with rational Schubert solutions, all well-defined transverse residues of the assembly are exactly 0, in the two ways the physics demands: 50/50 zeros on kissing-pentagon cuts (at  $n = 5$  all disjoint non-adjacent chords cross, so every transverse kissing cut is non-planar), and 20/20 zeros on the plane-intersection branches of pentagon-box cuts (where the chiral numerators are designed to vanish). The remaining chord-branch pentagon-box configurations at  $n = 5$  are composite strata (cut solutions pass through polygon vertices; naive transverse residues are ray-dependent, which our procedure detects automatically); their iterated-residue census is left open. The residue machinery is calibrated in-code on the one-loop box quadruple cut (residues exactly  $\mp 1$ , invariant under representative changes).

### 3 Discussion

Three comments. First, the result is a verification with an unexpected edge: the quasi-linear machinery of [1], developed because “the tedious triangulation is inevitable” in general, is never needed for two-loop MHV in the raw sign-flip charts at  $n \leq 6$  — the naive term-by-term positive part assembles into the exact integrand. Given the structural facts (i)–(ii), which hold at every  $n$ , it is natural to conjecture that (3) holds for all  $n$  at two loops MHV; we have no proof, and the composite leading-singularity sector remains unchecked beyond the families listed. Second, two predictions made during this work were refuted by its own tests and are recorded in the repository alongside the confirmations: an expectation of unit transverse residues on  $n = 5$  kissing cuts (planarity forces zeros instead), and the expectation that quasi-linearity would first appear at  $n = 5$  or 6 (it does not). Third, everything here is exact, small, and reproducible: the computations run in minutes on a laptop from the public repository, and we would welcome correction or extension by practitioners — in particular a proof or counterexample for the all- $n$  conjecture, and the composite-residue census.

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### References

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